

Step-by-Step RHEL 7.9 to RHEL 8.9 Upgrade Using Leapp Tool

1. Overview

Upgrading from RHEL 7.9 to RHEL 8.9 is a common task in modernizing infrastructure. The Leapp tool, developed by Red Hat, enables in-place upgrades safely and efficiently. This document provides a real-time, professional guide with all necessary steps, including visual diagrams and troubleshooting tips.

2. Real-World Scenario

Environment:

- Source: rhel7srv01 (RHEL 7.9)
- Target: In-place upgraded to RHEL 8.9
- Goal: Preserve user data, applications, and system configuration.

3. Prerequisites

- RHEL 7.9 system registered with Red Hat Subscription Manager.
- yum working with valid Red Hat repos.
- SELinux in enforcing or permissive mode.
- Complete system backup or VM snapshot.
- No third-party/unsupported kernel drivers.

4. Install Leapp Utility

Run the following command:

```
sudo yum install leapp leapp-repository -y
```

5. Pre-upgrade Check

Run the following command:

```
sudo leapp preupgrade
```

Output location:

- /var/log/leapp/leapp-report.txt
- /var/log/leapp/leapp-preupgrade.log

Example Inhibitor Warning:

Risk Factor: Inhibitor
Title: Unsupported driver detected: r8169
Solution: Replace driver or remove it before proceeding

Action:

- Remove or blacklist unsupported drivers
- Disable unnecessary third-party repos

6. Execute the Upgrade

Run the following command:

sudo leapp upgrade

7. Reboot Into Upgrade Environment

Run the following command:

sudo reboot

System will automatically boot into the Leapp-generated upgrade environment.

8. Post-upgrade Validation

Run the following commands:

cat /etc/redhat-release
uname -r

Restore SELinux, firewalld, and third-party repos as needed.

9. Diagram: Upgrade Workflow

```
+-----+
| Take Backup |
+-----+
|
v
+-----+
| Install Leapp |
+-----+
```

```

/
v
+-----+-----+
/ Pre-upgrade Check /
+-----+-----+
/
v
+-----+-----+
/ Fix Inhibitors /
+-----+-----+
/
v
+-----+-----+
/ Run Upgrade /
+-----+-----+
/
v
+-----+-----+
/ Reboot System /
+-----+-----+
/
v
+-----+-----+
/ Validate Upgrade /
+-----+-----+

```

10. Common Issues & Fixes

Issue	Resolution
Inhibitor: Unsupported Driver	Remove or replace with supported version
Subscription Errors	Run `subscription-manager register` and `attach`
Insufficient Space	Clean /boot, /var/cache/yum

11. Final Notes

- Always test upgrades in staging environments.
- Leapp does not support direct upgrade from CentOS 7 to CentOS Stream or RHEL 9.
- For CentOS 7, convert to RHEL before using Leapp.